Simplified RC4 Stream Cipher

Let us consider the stream cipher RC4, but instead of the full 256 bytes, we will use 8 x 3-bits. That is, the state vector **S** is 8 x 3-bits. We will operate on 3-bits of plaintext at a time since S can take the values 0 to 7, which can be represented as 3 bits.

Assume we use a 4 x 3-bit key of **K** = [1 2 3 6]. And a plaintext **P** = [1 2 2 2]

a. Initialize the state vector S and temporary vector T
b. Perform the initial permutation on S.

```
j = 0;
for i = 0 to 7 do
j = (j + S[i] + T[i]) mod 8
Swap(S[i],S[j]);
end
```

c. generate 3-bits at a time, k, that we XOR with each 3-bits of plaintext to produce the ciphertext. The 3-bits k is generated by:

```
i, j = 0;
while (true) {
      i = (i + 1) mod 8;
      j = (j + S[i]) mod 8;
      Swap (S[i], S[j]);
      t = (S[i] + S[j]) mod 8;
      k = S[t]; }
```

d. Calculate the Ciphertext C corresponding to the plaintext P

References

Simplified RC4 Example by Steven Gordon